When Does Disclosure Help? Information and Coordination in Equity Financing

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Introduction

US private firms don't have to publish or audit their financial statements, unlike in most other countries (Minnis and Shroff (2017)).

Policy debate: should private firms disclose more?

- 2017: SEC Commissioner Kara Stein asks whether disclosure laws should apply to firms that stay private.
- 2022: the SEC starts work on a plan requiring private firms to report regularly.

A disclosure mandate does more than inform investors: it makes information public.

In equity financing, investors' decisions are strategically linked: each cares about what the others know.

Research question: When does more information about firms raise welfare, and does it matter if it is public or private?

What I do

Propose a model where:

- Firms differ in productivity and raise capital from outside equity investors.
- Investors compete to fund firms but don't observe their productivity.
- Investment decision relies on private and public information about firm productivity.

I characterize:

- Equilibrium investment strategy of investors.
- The constrained social planner's solution.
- How welfare changes with the precision of public and private information.
- The optimal information composition (public vs private).

Model - Overview

- Two periods. General Equilibrium.
- Three agents: Firms, Investors, Household.

Firms: Measure one indexed by $j \in J = [0, 1]$.

- Differ in their productivity z_j and produce using capital only.
- Raises equity funding from a continuum of investors.
- Investors receive a fraction of firm profits.

Investors: Measure one indexed by $i \in I = [0, 1]$.

- Compete to provide capital to firms.
- Don't observe firms' productivity. Receive a private and a public signal about it.

Household:

- Risk Neutral. Extract utility from consumption only.
- Save by providing capital to investors.

Firms

Firm j raises equity funding $k_{i,j}$ from a continuum of investors $i \in I$.

Effective capital (CES):

$$K_j = \left[\int k_{i,j}^{\rho} \mathrm{d}i \right]^{1/\rho}, \quad \rho \in (0, 1]$$

Output is given by: $z_j K_j^{\alpha}$

Firms

Equity share: investors receive fraction λ of the firm's profit; founder keeps $1 - \lambda$.

$$\text{Total payout to investors} = \lambda \cdot z_j K_j^{\alpha}$$

Firms' Problem: Firm j chooses λ that maximizes founder's share of profits:

$$\max_{\lambda} (1 - \lambda) \cdot z_j K_j(\lambda)^{\alpha}$$

Note: Firm takes investors' response to λ into account when choosing it.

Profit Sharing Rule

A firm allocates the payout pool across investors using a sharing rule: $s_{i,j} = s(k_{i,j}, k_{-i,j})$.

$$\text{Investor } i \text{ payoff} = s_{i,j} \cdot \lambda z_j K_j^{\alpha}$$

Assumptions:

- Feasibility: $\int_I s_{i,j} \mathrm{d}i = 1$ and $s_{i,j} \geq 0$.
- Scale invariance: $s(\lambda k_{i,j}, \lambda k_{-i,j}) = s(k_{i,j}, k_{-i,j})$.
- Independence / separability: $s_{i,j} = \dfrac{f(k_{i,j})}{\int_I f(k_{\ell,j}) \mathrm{d}\ell}$, f increasing and continuous.

Any sharing rule satisfying these properties must be of the form (Skaperdas (1996)):

$$s_{i,j} = \frac{k_{i,j}^{\gamma}}{\int_I k_{\ell,j}^{\gamma} \mathrm{d}\ell}.$$

I will further impose diminishing returns: $\gamma \in (0, 1)$. Taste Shocks Motivation

Information Structure

Productivity follows a log-normal distribution:

$$\ln z_j \sim \mathcal{N}(\mu_z, 1/\kappa_z)$$

Each investor i observes:

- A private signal ($x_{i,j}$) about the productivity of firm j:

$$x_{i,j} = \ln z_j + \nu_{i,j}, \qquad \nu_{i,j} \overset{iid}{\sim} \mathcal{N}(0, 1/\kappa_x)$$

- A public signal (y_j) about the productivity of firm j.

$$y_j = \ln z_j + \varepsilon_j, \quad \varepsilon_j \overset{iid}{\sim} \mathcal{N}(0, 1/\kappa_y)$$

$\omega_{i,j} = (x_{i,j}, y_j) \in \Omega$: investor i's information set about firm j.

Investor Problem

Investors don't observe z_j.

They decide how much capital to provide to maximize expected profit:

$$\max_k \mathbb{E}\left[\frac{k^\gamma}{\int_I k_{i,j}^\gamma \mathrm{d}i}\lambda z_j K_j^\alpha - (1-\tau)Rk \mid \omega_{i,j}\right]$$

where:

- R is the cost of capital to the investor.
- τ is a tax/subsidy on the cost of capital.

Equilibrium Characterization

Investors play a Bayesian game: each investor i chooses k after observing $\omega_{i,j}$.

The symmetric Bayes–Nash equilibrium is a strategy $k : \Omega \to \mathbb{R}_+$ satisfying the fixed point below:

$$k(\omega_{i,j})^{1-\gamma} = \frac{\gamma\lambda}{R\,(1-\tau)}\mathbb{E}\left[\frac{z_j K_j^{\alpha}}{\int_I k(\omega_{i,j})^{\gamma}\mathrm{d}i} \mid \omega_{i,j}\right], \quad \forall \omega_{i,j} \in \Omega$$

where $K_j = \left[\int_I k(\omega_{i,j})^{\rho}\mathrm{d}i\right]^{1/\rho}$

Proposition 1 (Equilibrium Strategy). The equilibrium strategy $k(\cdot)$ is log-linear in the private and public signals:

$$\ln k(\omega_{i,j}) = \phi_0 + \phi_x x_{i,j} + \phi_y y_j$$

where:

$$\phi_x = \frac{1}{1-\alpha} \cdot \frac{(1-\theta)\,\kappa_x}{\kappa_z + \kappa_y + (1-\theta)\,\kappa_x}$$

$$\phi_y = \frac{1}{1-\alpha} \cdot \frac{\kappa_y}{\kappa_z + \kappa_y + (1-\theta)\,\kappa_x}$$

and $\theta = (\alpha - \gamma)/(1-\gamma)$ is the degree of strategic complementarity.

Expression for ϕ_0 Details θ

Given the equilibrium strategy, the solution to the firm's problem is:

$$\arg\max_{\lambda} (1-\lambda) \cdot z_j K_j(\lambda)^{\alpha} = \alpha$$

Households

- Risk neutral representative household.
- Endowed with e units of the consumption good in period 1.
- They save $(\bar{K})$ by supplying funds to investors at interest rate R.

Household problem:

$$\max_{c_1, c_2, \bar{K}} c_1 + \beta c_2$$

subject to:

$$c_1 + \bar{K} \leq e$$

$$c_2 + T \leq (1 - \delta + R)\, \bar{K} + \int_J \pi_j^F \, \mathrm{d}j + \int_I \pi_i^I \, \mathrm{d}i$$

T: lump-sum tax. π^F, π^I: firm and investor profits

Equilibrium

An equilibrium will consist of:

- A strategy $k : \Omega \to \mathbb{R}_+$ for investors.
- Consumption/savings: $c_1, c_2, \bar{K}$.
- Investors share of firm profits: λ.
- Price/Fiscal Policy: R and (τ, T).

Such that:

- $k(\cdot)$ is a Bayes–Nash equilibrium of the investors' game.
- c_1, c_2 and $\bar{K}$ solve the household problem.
- λ solves the firm's problem.
- Markets clear and government's budget constraint binds:

$$\int_I \int_J k(\omega_{i,j}) \mathrm{d}j \mathrm{d}i = \bar{K}$$

$$c_1 = e - \bar{K} \qquad c_2 = \int_J z_j \cdot K_j^{\alpha} \mathrm{d}j + (1-\delta)\bar{K}$$

$$\tau R \bar{K} = T$$

Constrained Social Planner Problem

The constrained social planner's problem is given by:

$$\max_{c_1, c_2, \{k_{i,j}\}_{i \in I, j \in J}} \mathbb{E}\left[c_1 + \beta c_2\right]$$

subject to:

- $c_1 + \int_I \int_J k_{i,j} \mathrm{d}j \mathrm{d}i = e$
- $c_2 = \int_J z_j K_j^{\alpha} \mathrm{d}j + (1-\delta) \int_I \int_J k_{i,j} \mathrm{d}j \mathrm{d}i$
- $k_{i,j}$ measurable w.r.t. $\omega_{i,j}$

The solution to the planner's problem satisfies the following fixed point:

$$k^S(\omega_{i,j})^{1-\rho} = \frac{\alpha\beta}{1-\beta(1-\delta)} \mathbb{E}\left[z_j K_j^{\alpha-\rho} \mid \omega_{i,j}\right], \quad \forall \omega_{i,j} \in \Omega$$

where $K_j = \left[\int_I k^S(\omega_{i,j})^{\rho} \mathrm{d}i\right]^{1/\rho}$

Constrained Social Planner Problem

Proposition 2 (Constrained Social Planner Solution). The solution of the constrained social planner's problem can be written as:

$$\ln k^S(\omega_{i,j}) = \phi_0^S + \phi_x^S x_{i,j} + \phi_y^S y_j$$

where:

$$\phi_x^S = \frac{1}{1-\alpha} \cdot \frac{(1-\theta^*)\,\kappa_x}{\kappa_z + \kappa_y + (1-\theta^*)\,\kappa_x}$$

$$\phi_y^S = \frac{1}{1-\alpha} \cdot \frac{\kappa_y}{\kappa_z + \kappa_y + (1-\theta^*)\,\kappa_x}$$

and $\theta^* = \frac{\alpha-\rho}{1-\rho}$ is the socially optimal degree of coordination.

Constrained Social Planner Problem

Proposition 3 (Efficiency Conditions). The constrained social planner's solution coincides with the decentralized equilibrium if:

1. $\gamma = \rho \implies \theta = \theta^*$
2. $1 - \tau = \gamma \implies \phi_0 = \phi_0^S$ (given condition 1)

Intuition:

- Condition 1: Ensures that the degree of strategic complementarity in the decentralized equilibrium equals the socially optimal one. Why $\gamma = \rho$?

- Condition 2: Offsets investors' market power.

Welfare in the decentralized equilibrium

Welfare in the decentralized economy can be written as:

$$\mathcal{W} = e - [1 - \beta(1-\delta)] \cdot \bar{K} + \beta \cdot \mathcal{Y}, \qquad \mathcal{Y} = \int_J z_j K_j^{\alpha} \mathrm{d}j.$$

Lemma 1. In equilibrium, $\bar{K} = \chi \mathcal{Y}$ with $\chi = \frac{\gamma\alpha}{(1-\tau)R}$. Then:

$$\mathcal{W} = e + m\,\bar{K}, \qquad m = [1 - \beta(1-\delta)] \left(\frac{1-\tau}{\gamma\alpha} - 1 \right)$$

I can focus on the effect of information on $\bar{K}$ to study its effect on welfare.

Welfare rises with $\bar{K}$ unless the subsidy over-supplies capital.

- $\operatorname{sgn}(m) = \operatorname{sgn}(1 - \tau - \gamma\alpha)$.

Efficient Use of Information

Proposition 4 (Efficient Use of Information). Suppose $\gamma = \rho$ and $\tau = 1 - \gamma$. Then:

$$\frac{\partial \mathcal{W}}{\partial \kappa_x} > 0 \qquad \text{and} \qquad \frac{\partial \mathcal{W}}{\partial \kappa_y} > 0$$

Intuition:

- $\tau = 1 - \gamma$ makes $m > 0$.
- $\gamma = \rho$ makes investors use information efficiently, so more of either signal raises $\bar{K}$.

Counterpart of the efficient use of information result in Angeletos and Pavan (2007).

When does more public information raise capital?

Proposition 5 (Public Information and Aggregate Capital).
Define the threshold $\Delta_{\kappa_y} = \frac{(1-\alpha)(2\rho-\gamma-1)}{(1-\gamma)^2}$. Then:

$$\operatorname{sgn}\left(\frac{\partial \bar{K}}{\partial \kappa_y}\right) = \operatorname{sgn}\left(\frac{\kappa_y + \kappa_z}{\kappa_x} - \Delta_{\kappa_y}\right)$$

The sign is the same for aggregate output and TFP. Why?

Capital can fall with more public information when:

- Coordination motive is strong ($\rho > \frac{1+\gamma}{2}$).
- Public information is imprecise relative to private information ($\frac{\kappa_y+\kappa_z}{\kappa_x} < \Delta_{\kappa_y}$).

$\rho > \frac{1+\gamma}{2} \iff \theta > \frac{1+\theta^*}{2}$: investors coordinate too much.

When does more public information help?

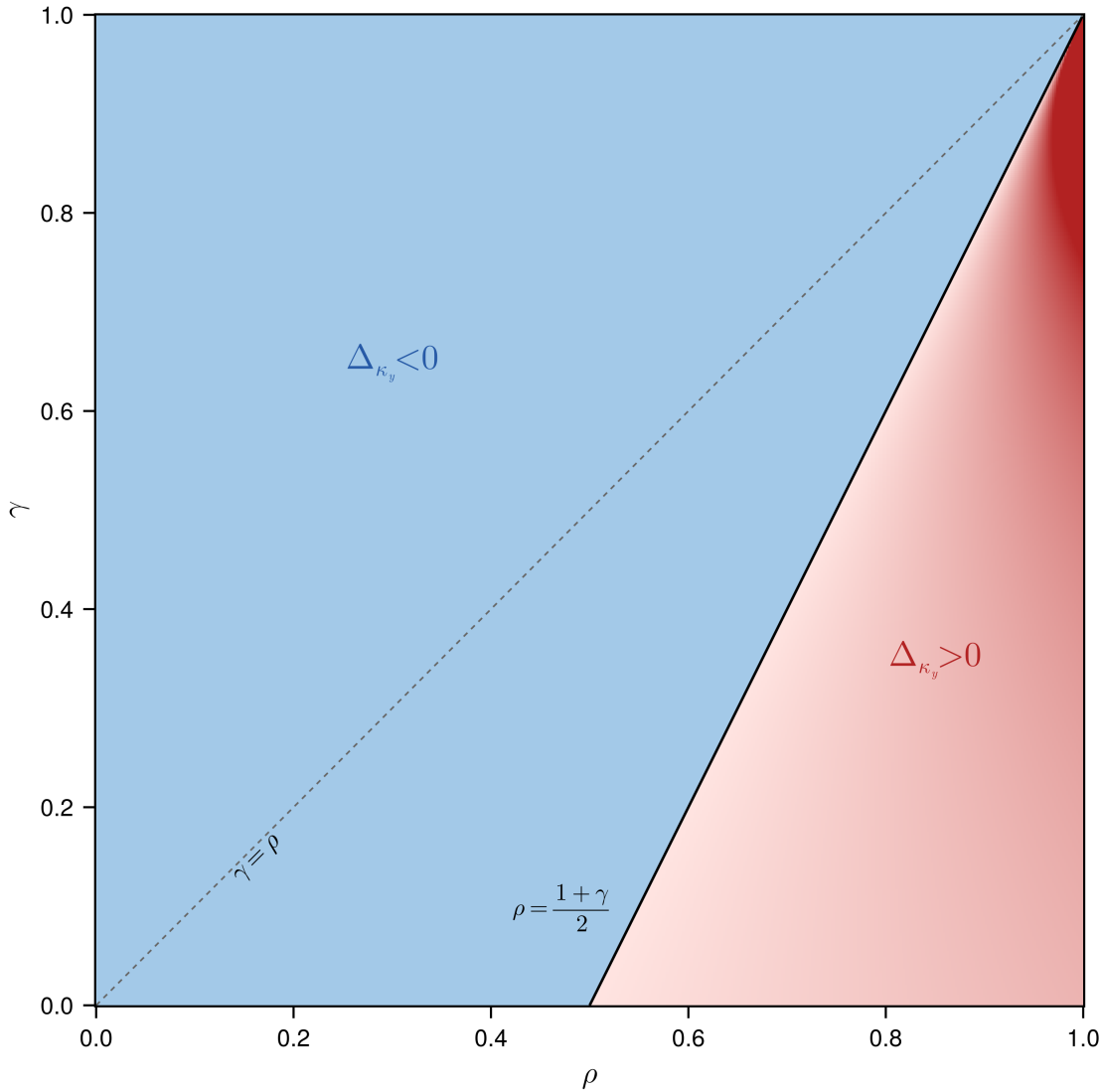
Corollary 1 (Public Information and Welfare). By Lemma 1, $\frac{\partial \mathcal{W}}{\partial \kappa_y} = m \frac{\partial \bar{K}}{\partial \kappa_y}$. Therefore:

$$\operatorname{sgn}\left(\frac{\partial \mathcal{W}}{\partial \kappa_y}\right) = \operatorname{sgn}(m) \cdot \operatorname{sgn}\left(\frac{\kappa_y + \kappa_z}{\kappa_x} - \Delta_{\kappa_y}\right)$$

When $m > 0$, welfare can fall with more public information exactly when capital does.

When $m < 0$, every sign reverses.

Illustration: Welfare Threshold



- Blue region: more public information always improves welfare.
- Red region: more public information can hurt.
- Darker shade $\Rightarrow$ higher threshold Δ_{κ_y}.

Regions shown for $m > 0$. Signs reverse when $m < 0$.

When does more private information raise capital?

Proposition 6 (Private Information and Aggregate Capital). Let $\rho < 1$ and define the threshold $\Delta_{\kappa_x} = \frac{(1-\gamma)(2\gamma-\rho-1)}{(1-\alpha)(1-\rho)}$. Then:

$$\operatorname{sgn}\left(\frac{\partial \bar{K}}{\partial \kappa_x}\right) = \operatorname{sgn}\left(\frac{\kappa_x}{\kappa_y + \kappa_z} - \Delta_{\kappa_x}\right)$$

The sign is the same for aggregate output and TFP. Why?

Capital can fall with more private information when:

- Coordination motive is weak ($\gamma > \frac{1+\rho}{2}$).
- Private information is imprecise relative to public information ($\frac{\kappa_x}{\kappa_y+\kappa_z} < \Delta_{\kappa_x}$).

$\gamma > \frac{1+\rho}{2} \iff \theta^* > \frac{1+\theta}{2}$: investors coordinate too little.

When does more private information help?

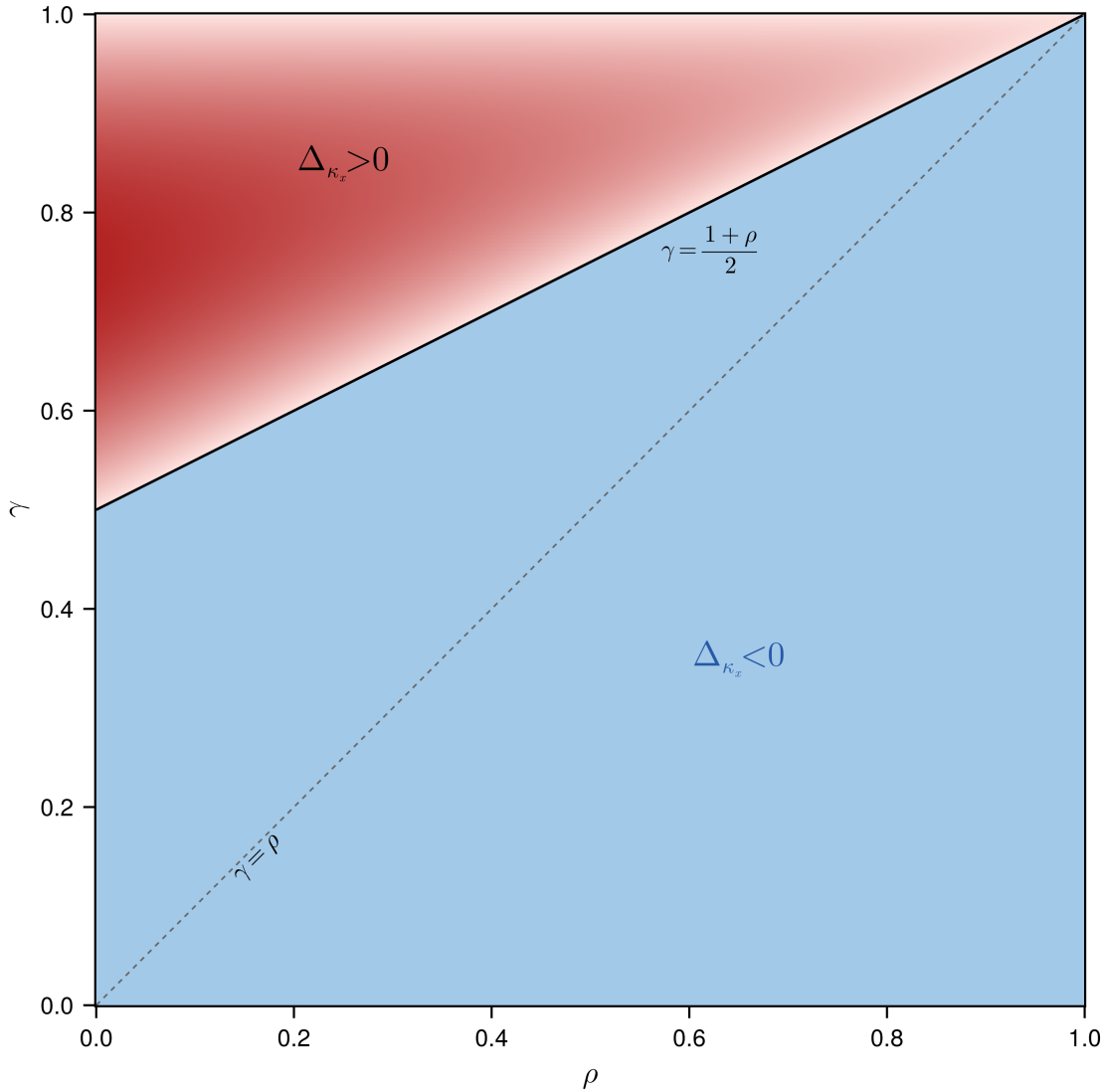
Corollary 2 (Private Information and Welfare). By Lemma 1, $\frac{\partial \mathcal{W}}{\partial \kappa_x} = m \frac{\partial \bar{K}}{\partial \kappa_x}$. Therefore:

$$\operatorname{sgn}\left(\frac{\partial \mathcal{W}}{\partial \kappa_x}\right) = \operatorname{sgn}(m) \cdot \operatorname{sgn}\left(\frac{\kappa_x}{\kappa_y + \kappa_z} - \Delta_{\kappa_x}\right)$$

When $m > 0$, welfare can fall with more private information exactly when capital does.

When $m < 0$, every sign reverses.

Illustration: Welfare Threshold



- Blue region: more private information always improves welfare.
- Red region: more private information can hurt.
- Darker shade $\Rightarrow$ higher threshold Δ_{κ_x}.

Regions shown for $m > 0$. Signs reverse when $m < 0$.

When Should Information Be Public?

Fix $\kappa = \kappa_x + \kappa_y$.

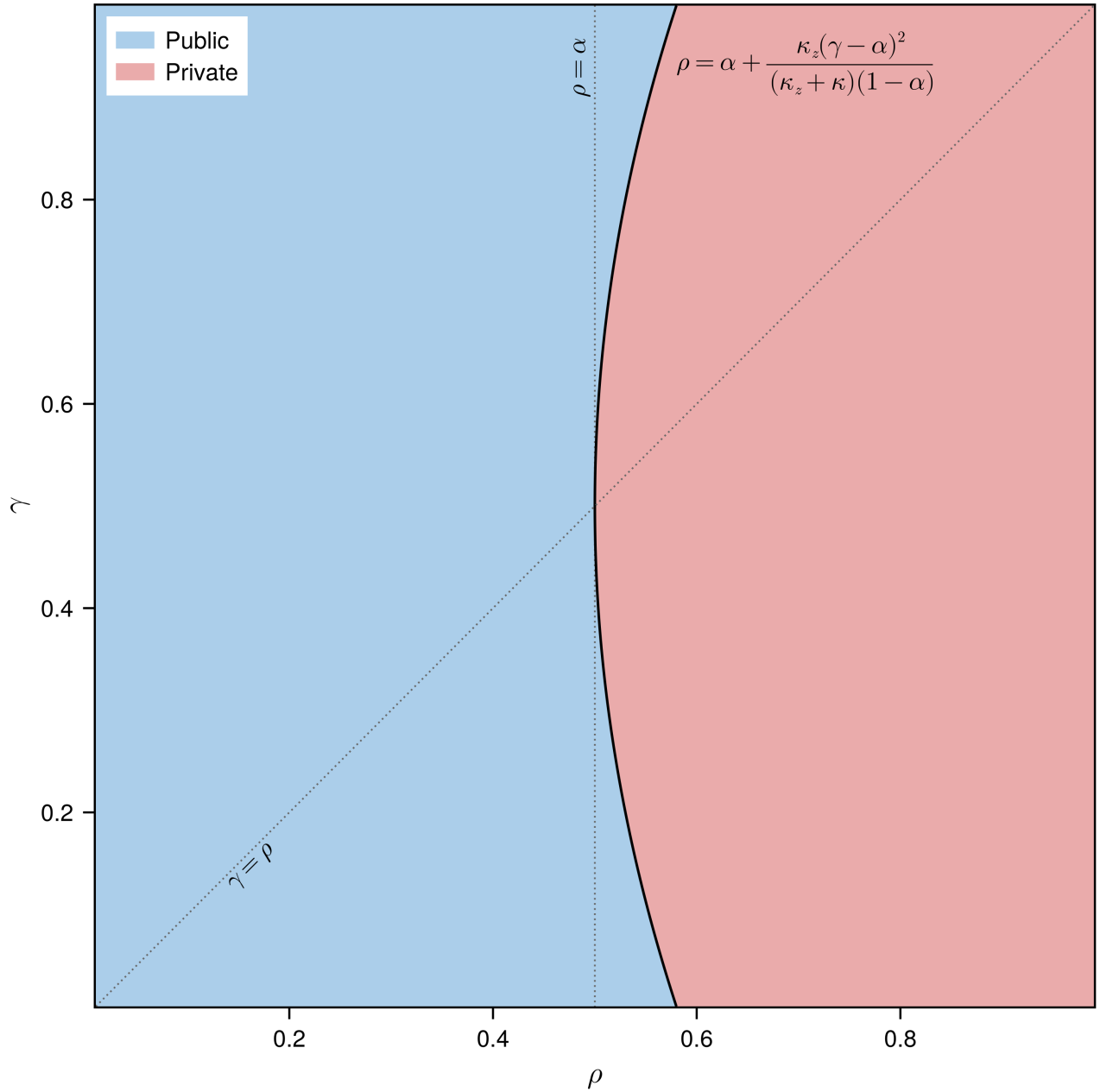
How to split κ between private (κ_x) and public (κ_y) information to maximize welfare in the decentralized equilibrium?

Proposition 7 (Optimal Information Structure). Let $m > 0$. The welfare-maximizing information structure is either fully public ($\kappa_y = \kappa$) or fully private ($\kappa_x = \kappa$).

Information should be fully public if and only if:

$$\rho < \alpha + \frac{\kappa_z \left(\gamma - \alpha\right)^2}{\left(\kappa_z + \kappa\right)\left(1 - \alpha\right)}$$

When Should Information Be Public?



Detailed Plot

Conclusion

- I study when improving investors' information about firm productivity raises welfare in equity financing.

- Larger contributions by other investors raise the payout, but dilute each investor's share of it.

- The balance between these forces determines whether investors react to one another too much or too little.

- More information can be welfare reducing!

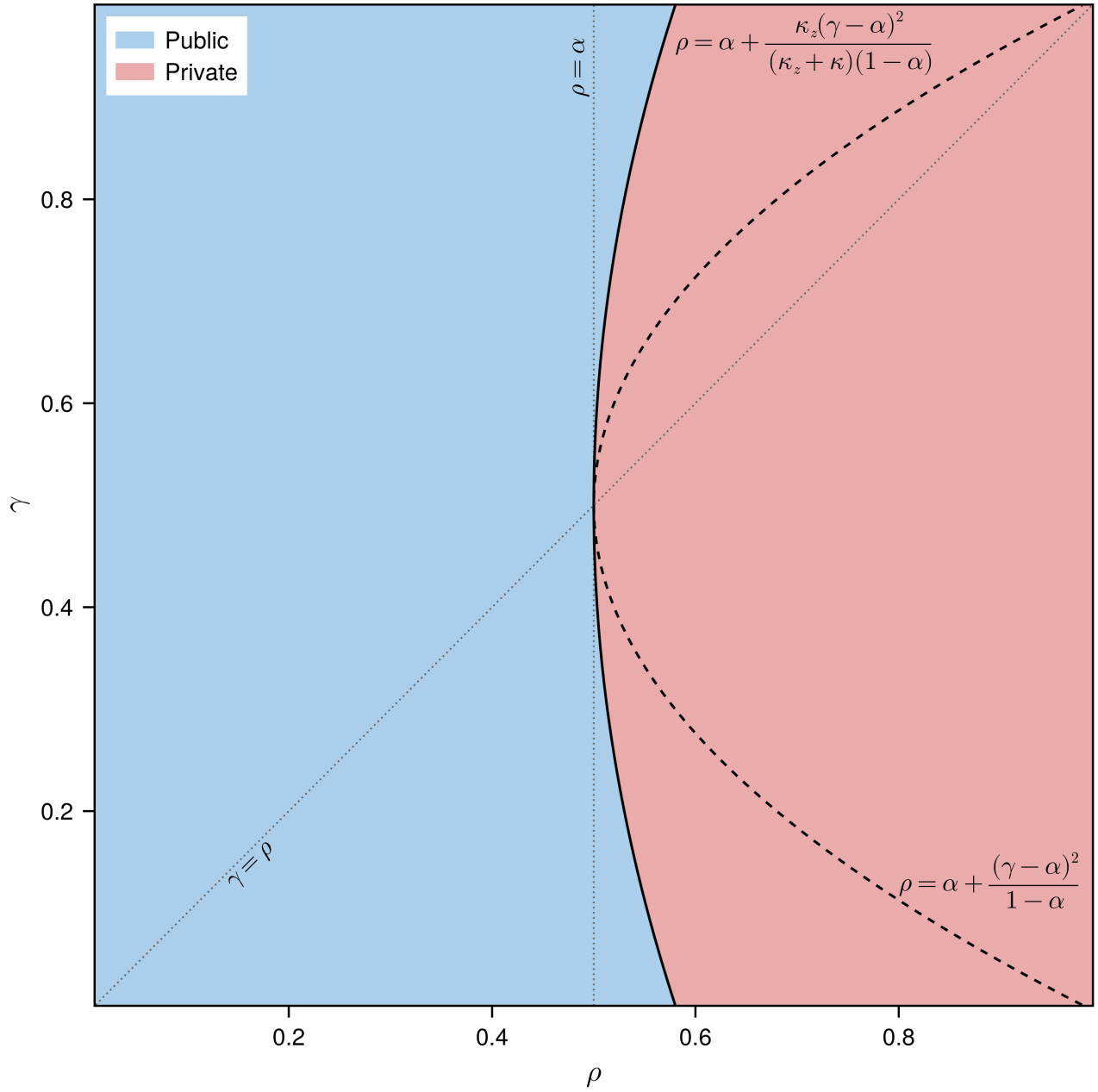
    - Public information can hurt when coordination motives are strong and public information is relatively imprecise.

    - Private information can hurt when coordination motives are weak and private information is relatively imprecise.

- Optimal information composition is either fully public or fully private.

Appendix

When Should Information Be Public?



Extension - Relaxing Risk Neutrality

Assume that HH has utility function $U(c_1, c_2)$, increasing and concave.

If $1 - \tau = \gamma$, then the welfare results holds for any U.

- The tax sets $\bar{K}$ at its optimal level $\implies$ only TFP channel survives.
- The TFP channel is independent of HH's preferences.

Recall result for TFP:

Proposition 8 (Effect of κ_y on TFP). Let $\Delta_{\kappa_y} = \frac{(1-\theta)(2\theta-1-\theta^*)}{1-\theta^*}$ as in the original model. Then:

$$\frac{\partial \text{TFP}}{\partial \kappa_y} > 0 \iff \frac{\kappa_y + \kappa_z}{\kappa_x} > \Delta_{\kappa_y}$$

Extension - Endogenous Disclosure

At $t = 0$, before productivity is realized, firm decides how much public information to disclose.

They will choose the precision of the public signal (κ_y) facing a cost $C(\kappa_y)$ to be paid at $t = 1$.

Then the game unfolds as before.

Firm's problem at $t = 0$:

$$\max_{\kappa_y} \beta \cdot (1 - \alpha) \underbrace{\mathbb{E}\left[z_j K_j^{\alpha}\right]}_{\mathcal{Y}} - C(\kappa_y)$$

where $\mathcal{Y}$ is aggregate output.

Extension - Endogenous Disclosure

In an interior solution, the first order condition is:

$$\beta(1-\alpha)\cdot\frac{\partial\mathcal{Y}}{\partial\kappa_y}=C'(\kappa_y^F)$$

where $\frac{\partial\mathcal{Y}}{\partial\kappa_y}$ was already characterized in the baseline model.

Now, consider what a planner would choose for κ_y at $t=0$ to maximize welfare in the decentralized equilibrium:

$$\max_{\kappa_y}\mathbb{E}\left[e-(1-\beta)\bar{K}+\beta z_j K_j^{\alpha}\right]-C(\kappa_y)$$

Extension - Endogenous Disclosure

We can rewrite the planner's problem as:

$$\max_{\kappa_y} e + \beta \left(1 - \frac{\alpha\gamma}{1-\tau}\right) \mathcal{Y} - C\left(\kappa_y\right)$$

First order condition:

$$\beta \left(1 - \frac{\alpha\gamma}{1-\tau}\right) \cdot \frac{\partial \mathcal{Y}}{\partial \kappa_y} = C'(\kappa_y^P)$$

Proposition 9 (Endogenous Disclosure). In an interior solution:

- If $\tau < 1 - \gamma$, then $\kappa_y^F < \kappa_y^P$. (Under-disclose)
- If $\tau = 1 - \gamma$, then $\kappa_y^F = \kappa_y^P$. (Socially optimal disclosure)

Back

Taste Shocks - Motivation

Assume that firm j owner assigns a perceived value $V_{i,j}$ when matching with investor i given by:

$$V_{i,j} = \eta \cdot f(k_{i,j}) + \varepsilon_{i,j}, \quad \varepsilon_{i,j} \overset{iid}{\sim} \text{Extreme Value Type I}$$

- η is the weight the founder places on capital size.
- $\varepsilon_{i,j}$ is an idiosyncratic preference shock.
- $f' > 0$

Intuition:

- The founder wants to give more shares to those investors who provide more capital, but the process is noisy due to frictions.
- Frictions: personal relationship, marketing, search costs, who arrives first, etc.

Taste Shocks - Motivation

The share investor i gets is given by the logit demand:

$$s_{i,j} = \frac{\exp\left(\eta f(k_{i,j})\right)}{\int_I \exp\left(\eta f(k_{i,j})\right) \mathrm{d}i}$$

If we impose scale invariance, f must be of the form:

$$f(k_{i,j}) = a \cdot \ln(k_{i,j}) + b, \quad a > 0, b \in \mathbb{R}$$

Then the sharing rule becomes:

$$s_{i,j} = \frac{k_{i,j}^{\eta a}}{\int_I k_{i,j}^{\eta a} \mathrm{d}i}$$

Setting $\gamma = \eta a$ recovers the sharing rule in the main model.

Back

Defining Strategic Complementarity

I will define it as the log-derivative of the best response with respect to a proportional increase in others' choices.

Consider a proportional increase in others' choices by a factor t.

$$
\begin{aligned}
k_{i,j}^{1-\gamma} &= \frac{\gamma\lambda}{R\left(1-\tau\right)}\mathbb{E}\left[\frac{z_j\left[\int_I\left(tk_{i,j}\right)^{\rho}\mathrm{d}i\right]^{\alpha/\rho}}{\int_I\left(tk_{i,j}\right)^{\gamma}\mathrm{d}i}\mid\omega_{i,j}\right] \\
&= t^{\alpha-\gamma}\frac{\gamma\lambda}{R\left(1-\tau\right)}\mathbb{E}\left[\frac{z_j\left[\int_I k_{i,j}^{\rho}\mathrm{d}i\right]^{\alpha/\rho}}{\int_I k_{i,j}^{\gamma}\mathrm{d}i}\mid\omega_{i,j}\right]
\end{aligned}
$$

Then define the degree of strategic complementarity θ as:

$$
\theta = \frac{\partial\ln\left(k_{i,j}\right)}{\partial\ln\left(t\right)} = \frac{\alpha-\gamma}{1-\gamma}
$$

Back

Expression for ϕ_0

The expression for ϕ_0 is given by:

$$\phi_0 = \frac{1}{(1-\alpha)} \ln\left(\frac{\gamma\lambda}{R\,(1-\tau)}\right) + \frac{\kappa_z \mu_z}{(1-\alpha)\,D} + \frac{\kappa_x\left(\alpha\rho - \gamma^2\right) + (1-\gamma)^2\left(\kappa_x + \kappa_y + \kappa_z\right)}{2\,(1-\alpha)\,(1-\gamma)^2\,D^2}$$

where: $D = \kappa_y + \kappa_z + (1-\theta)\,\kappa_x$

Back

Alternative Environment

- Investors supply differentiated capital varieties.
- Firm j aggregates them with CES technology and rents each variety at price $P_{i,j}$.
- Investors compete in quantity.

Firm Problem: Firm j chooses $k_{i,j}$ to maximize profits:

$$\max_{k_{i,j}} z_j \left[\int_I k_{i,j}^{\rho} \mathrm{d}i\right]^{\alpha/\rho} - \int_I P_{i,j} k_{i,j} \mathrm{d}i$$

FOC:

$$P_{i,j} = \alpha z_j K_j^{\alpha-\rho} k_{i,j}^{\rho-1}$$

where $K_j = \left[\int_I k_{i,j}^{\rho} \mathrm{d}i\right]^{1/\rho}$.

Alternative Environment

Investor Problem: Investor i chooses $k_{i,j}$ to maximize expected profits:

$$\max_{k_{i,j}} \mathbb{E}\left[P_{i,j}k_{i,j} - R\left(1-\tau\right)k_{i,j} \mid \omega_{i,j}\right]$$

subject to: $P_{i,j} = \alpha z_j K_j^{\alpha-\rho} k_{i,j}^{\rho-1}$

Substituting the price into the investor's problem:

$$\begin{aligned}
\max_{k_{i,j}} \mathbb{E}\left[\alpha z_j K_j^{\alpha-\rho} k_{i,j}^{\rho-1} k_{i,j} - R\left(1-\tau\right)k_{i,j} \mid \omega_{i,j}\right] &= \max_{k_{i,j}} \mathbb{E}\left[\alpha z_j K_j^{\alpha-\rho} k_{i,j}^{\rho} - R\left(1-\tau\right)k_{i,j} \mid \omega_{i,j}\right] \\
&= \max_{k_{i,j}} \mathbb{E}\left[\alpha z_j K_j^{\alpha} \frac{k_{i,j}^{\rho}}{K_j^{\rho}} - R\left(1-\tau\right)k_{i,j} \mid \omega_{i,j}\right] \\
&= \max_{k_{i,j}} \mathbb{E}\left[\alpha z_j K_j^{\alpha} \frac{k_{i,j}^{\rho}}{\int_I k_{m,j}^{\rho} \mathrm{d}m} - R\left(1-\tau\right)k_{i,j} \mid \omega_{i,j}\right]
\end{aligned}$$

Alternative Environment

Last line of the previous frame can be interpreted as:

$$\max_{k_{i,j}} \mathbb{E}\left[\underbrace{\alpha}_{1}\underbrace{z_j K_j^{\alpha}}_{2}\underbrace{\frac{k_{i,j}^{\rho}}{\int_I k_{m,j}^{\rho}\mathrm{d}m}}_{3} - R\left(1-\tau\right)k_{i,j} \mid \omega_{i,j}\right]$$

1. α: share of profits that investors get. Firm's problem $\implies \lambda = \alpha$
2. $z_j K_j^{\alpha}$: firm j's output/profits
3. Sharing rule: $s_{i,j} = \dfrac{k_{i,j}^{\rho}}{\int_I k_{m,j}^{\rho}\mathrm{d}m}$ when $\gamma = \rho$.

Monopolistic competition in differentiated capital is isomorphic to the original model with $\gamma = \rho$.

Back

Capital, Output and TFP

By Lemma 1, output is proportional to capital:

$$\mathcal{Y} = \frac{\bar{K}}{\chi}, \qquad \chi = \frac{\gamma\alpha}{(1-\tau)R}$$

Writing $\mathcal{Y} = \mathsf{TFP} \cdot \bar{K}^{\alpha}$:

$$\mathsf{TFP} = \frac{\bar{K}^{1-\alpha}}{\chi}$$

χ does not depend on κ_x or κ_y. Therefore, for $\kappa \in \{\kappa_x, \kappa_y\}$:

$$\operatorname{sgn}\left(\frac{\partial \bar{K}}{\partial \kappa}\right) = \operatorname{sgn}\left(\frac{\partial \mathcal{Y}}{\partial \kappa}\right) = \operatorname{sgn}\left(\frac{\partial \mathsf{TFP}}{\partial \kappa}\right)$$

Back to κ_y Back to κ_x